

CS2422 Assembly Language & System Programming

October 2, 2003

Today's Topics

- Data-Related Operators (OFFSET, PTR, ...etc.)

Study Guide

- Section 4.3: Data-Related Operators

Data Related Operators

- Who are they?
 - OFFSET, PTR, TYPE, LENGTHOF, SIZEOF
- They are only understood by the assembler.
- They are not instructions! For example:
`MOV EDI, OFFSET Var1`

Operand Sizes

- Operands may have the size of 1 byte, 2 bytes, or 4 bytes.
- Most of time, we can tell the size from the register names or the variable definition.
For examples:

```
Var1    BYTE "Hello"  
        MOV ECX, 13  
        MOV AL, Var1
```

PTR

- But sometimes we cannot tell the size of operand, especially if indirect addressing (or pointer) is used.
- Or we may simply want to override the default.

Some examples in next slide...

```
myDouble    DWORD    12345678h
    MOV AL, myDouble ; error
    MOV AL, BYTE PTR myDouble
    MOV AX, WORD PTR myDouble
    MOV AX, WORD PTR [myDouble+2]
    MOV EAX, myDouble
```

What Else?

- TYPE returns the size (in bytes) of each element.
- LENGTHOF returns the number of elements.
- SIZEOF returns the size of the variable (the whole array).
 - ➔ $\text{SIZEOF} = \text{LENGTHOF} * \text{TYPE}$

I lied when I said there was no array data type in assembly...

.data

byte1 BYTE 10, 20, 30

array1 WORD 30 DUP(?)

- Exercise: What is TYPE byte1? TYPE array1?
- LENGTHOF array1 is 30, SIZEOF array1 is 60.

What Have We Learned So Far?

A bird's-eye view:

- It's a very different world from high-level languages.
- It's important to access the data (in memory) precisely at our will.
- Flags to control the execution flow.

Where Do We Go from Here?

- Conditional Branches (similar to if...then...)
- And most importantly...

Get a feeling of how the low-level actions in assembly level become the fancy Windows (or Linux) operating system and applications.